

### **Continuous Probability Distribution:**

The probability distribution of continuous random variable that arises from some special experiments is known as continuous probability distribution.

The function  $f(x)$  is a probability density function (pdf) for the continuous random variable  $X$ , defined over the set of real numbers, if

1.  $f(x) \geq 0$ , for all real numbers,  $x \in R$

2.  $\int_{-\infty}^{\infty} f(x) dx = 1$

3.  $P(a < x < b) = \int_a^b f(x) dx$

The following are some of the commonly discussed continuous distribution:

- a) Normal Distribution
- b) Gamma Distribution
- c) Uniform Distribution
- d) Exponential Distribution

**Normal Distribution:**

Let us consider X is a normal variate with the two parameters mean  $\mu$  and variance  $\sigma^2$  then the probability density function of X is –

$$f(X) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2} ; \quad -\infty < x < \infty, -\infty < \mu < \infty \text{ and } \sigma > 0$$

Where,  $\sigma$  = standard deviation of x

$\mu$  = mean of x

$\pi = 3.1416$ ,  $e = 2.71828$ ;

**Standard Normal Distribution:**

When,  $\mu = 0$  and  $\sigma = 1$  then the Normal Distribution is called Standard Normal Distribution.

Let, X is a Normally distributed random variable with mean =  $\mu$  and variance =  $\sigma^2$ .

Then,  $Z = \frac{X - \mu}{\sigma}$ .

Where Z follows standard normal distribution.

**Empirical rule:**

$\mu \pm 1\sigma$  contain 68.25% observation.

$\mu \pm 2\sigma$  contain 95.44% observation.

$\mu \pm 3\sigma$  contain 99.73% observation.

**Example 1:**

The lifetime of a single electric bulb have a mean of 720 hours and standard deviation 10 hours. We are assuming that the distribution of lifetime measured is normal. Compute the percentage of bulbs that have lifetime more than 740 hours.

**Solution:**

Let, x = Lifetime of a electric bulb.

The mean lifetime  $\mu = 720$  hours and standard deviation,  $\sigma = 10$  hours

Now,

The probability that the batteries have lifetime more than 740 hours =  $P(X > 740)$

$$\begin{aligned} P(X > 740) &= P\left(\frac{X - \mu}{\sigma} > \frac{740 - \mu}{\sigma}\right) \\ &= P\left(Z > \frac{740 - 720}{10}\right) \\ &= P(Z > 2) \\ &= 1 - P(Z < 2) \\ &= 1 - 0.9772 \text{ [from table]} \\ &= 0.0228 \end{aligned}$$

So, the percentage of batteries that have lifetime more than 740 hours is  $(0.0228 \times 100) = 2.28\%$

**Example 2:**

If the height of 300 students are normally distributed with mean 68.0 inches and standard deviation 3.0 inches, how many students have height, (i) greater than 72 inches (ii) at least 65 inches (iii) less than or equal to 64 inches (iv) at most 80 inches (v) between 60 inches to 65 inches

(i) above 65 inches and less than 71 inches (vii) more than 74 inch but smaller than 80 inch (Viii) equal to 70 inches.

Solution:

Let,  $x$  = height of the students

$X$  is normally distributed with mean  $\mu = 68$  inches and standard deviation  $\sigma = 3$

Total number of students  $N = 300$

(i) Probability that the students have height greater than 72 inches  $= P(x > 72)$

$$\begin{aligned} P(x > 72) &= P\left(\frac{X-\mu}{\sigma} > \frac{72-\mu}{\sigma}\right) \\ &= P\left(z > \frac{72-68}{3}\right) \\ &= P(z > 1.33) \\ &= 1 - P(z < 1.33) \\ &= 1 - 0.9082 \\ &= 0.0918 \end{aligned}$$

The expected number of student  $= Np = 300 \times 0.0918 = 27.54 = 28$  (app)

(ii) Probability that the students have height at least 65 inches  $= P(x > 65)$

$$\begin{aligned} P(x > 65) &= P\left(\frac{X-\mu}{\sigma} > \frac{65-\mu}{\sigma}\right) \\ &= P\left(z > \frac{65-68}{3}\right) \\ &= P(z > -1) \\ &= 1 - P(z < -1) \\ &= 1 - 0.1587 \\ &= 0.8413 \end{aligned}$$

The expected number of student  $= Np = 300 \times 0.8413 = 252.39 = 252$  (app)

(iii) Probability that the students have height less than or equal to 64 inches  $= P(x \leq 64)$

$$\begin{aligned} P(x \leq 64) &= P\left(\frac{X-\mu}{\sigma} \leq \frac{64-\mu}{\sigma}\right) \\ &= P\left(z \leq \frac{64-68}{3}\right) \end{aligned}$$

$$= P(z \leq -1.33)$$

$$= 0.0918$$

The expected number of student =  $300 \times 0.0918 = 27.54 = 28$  (app)

(iv) Probability that the students have height at most 80 inches =  $P(x \leq 80)$

$$P(x \leq 80) = P\left(\frac{X-\mu}{\sigma} \leq \frac{64-\mu}{\sigma}\right)$$

$$= P\left(z \leq \frac{80-68}{3}\right)$$

$$= P(z \leq 4)$$

$$= 1$$

The expected number of student =  $300 \times 1 = 300$

(v) Probability that the students have height 60 inches to 65 inches

$$= P(60 < x < 65)$$

$$P(60 < x < 65) = P\left(\frac{60-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{65-\mu}{\sigma}\right)$$

$$= P\left(\frac{60-68}{3} < z < \frac{65-68}{3}\right)$$

$$= P(-2.67 < z < -1)$$

$$= P(z < -1) - P(z < -2.67)$$

$$= 0.1587 - 0.0038$$

$$= 0.1549$$

The expected number of student =  $300 \times 0.1549 = 46.47 = 46$  (app)

(vi) Probability that the students have height above 65 inches and less than 71 inches

$$= P(65 < x < 71)$$

$$P(65 < x < 71) = P\left(\frac{65-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{71-\mu}{\sigma}\right)$$

$$= P\left(\frac{65-68}{3} < z < \frac{71-68}{3}\right)$$

$$= P(-1 < z < 1)$$

$$= P(z < 1) - P(z < -1)$$

$$= 0.8413 - 0.1587$$

$$= 0.6826$$

The expected number of student =  $300 \times 0.6826 = 204.78 = 205$  (app)

(vii) Probability that the students have height more than 74 inch but smaller than 80 inch

$$= P(74 < x < 80)$$

$$\begin{aligned} P(74 < x < 80) &= P\left(\frac{74-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \frac{80-\mu}{\sigma}\right) \\ &= P\left(\frac{74-68}{3} < z < \frac{80-68}{3}\right) \\ &= P(2 < z < 4) \\ &= P(z < 4) - P(z < 2) \\ &= 1 - 0.9772 \\ &= 0.0228 \end{aligned}$$

The expected number of student =  $300 \times 0.0228 = 6.84 = 7$  (app)

(viii) Probability that the students have height equal to 70 inches =  $P(x = 70)$

$$P(x = 70) = 0$$

The expected number of students =  $300 \times 0 = 0$

# Normal Distribution

**Example**  
**4b**

If  $X$  is a normal random variable with parameters  $\mu = 3$  and  $\sigma^2 = 9$ , find  
(a)  $P\{2 < X < 5\}$ ; (b)  $P\{X > 0\}$ ; (c)  $P\{|X - 3| > 6\}$ .

**Solution** (a)

$$\begin{aligned}P\{2 < X < 5\} &= P\left\{\frac{2-3}{3} < \frac{X-3}{3} < \frac{5-3}{3}\right\} \\&= P\left\{-\frac{1}{3} < Z < \frac{2}{3}\right\} \\&= \Phi\left(\frac{2}{3}\right) - \Phi\left(-\frac{1}{3}\right) \\&= \Phi\left(\frac{2}{3}\right) - \left[1 - \Phi\left(\frac{1}{3}\right)\right] \approx .3779\end{aligned}$$

(b)

$$\begin{aligned}P\{X > 0\} &= P\left\{\frac{X-3}{3} > \frac{0-3}{3}\right\} = P\{Z > -1\} \\&= 1 - \Phi(-1) \\&= \Phi(1) \\&\approx .8413\end{aligned}$$

(c)

$$\begin{aligned}P\{|X - 3| > 6\} &= P\{X > 9\} + P\{X < -3\} \\&= P\left\{\frac{X-3}{3} > \frac{9-3}{3}\right\} + P\left\{\frac{X-3}{3} < \frac{-3-3}{3}\right\} \\&= P\{Z > 2\} + P\{Z < -2\} \\&= 1 - \Phi(2) + \Phi(-2) \\&= 2[1 - \Phi(2)] \\&\approx .0456\end{aligned}$$

